

A Blockchain-Anchored Digital Twin Framework with LSTM Forecasting and PPO Optimization for Safe ICU Ventilator Parameter Recommendation

Rishav Kumar, Shaik Md Althaf, Sohan M Raju, Gnyan Mallaiah
Department of Information Science and Engineering
RV College of Engineering, Bengaluru, India
{rishavkumar, smdalthaf, sohanmraju, gnyanmallaiah}.is22@rvce.edu.in

Prof. Chethana R (Guide)
Department of Information Science and Engineering
RV College of Engineering, Bengaluru, India
chethanar@rvce.edu.in

Abstract—Mechanical ventilation in the intensive care unit (ICU) involves continuous, high-stakes adjustment of coupled parameters (PEEP, FiO₂, tidal volume) under non-stationary patient physiology. Manual settings are infrequently revised and seldom personalized, while purely autonomous controllers conflict with the safety, traceability, and explainability demands of clinical practice. We propose a four-layer decision-support framework that couples (i) a calibrated patient-specific *Digital Twin* that simulates short-horizon respiratory dynamics, (ii) an *LSTM* forecasting engine that predicts near-future SpO₂ and hypoxia risk, (iii) a *Proximal Policy Optimization* (PPO) agent that searches a constrained action space inside the twin sandbox, and (iv) a *permissioned hash-chained audit ledger* that immutably records every recommendation, override, and acceptance. A clinician-in-the-loop guardrail clamps every action against ARDSNet-style safety bounds before it is offered. The system is implemented as a reproducible monorepo with a profile-driven synthetic ventilator simulator (normal, ARDS, COPD, unstable), a FastAPI orchestration surface, and a continuous-integration quality gate that enforces deterministic-replay and trend-direction thresholds on every commit. On the current Phase 2 benchmark across six replay scenarios the calibrated twin achieves 100% trend-direction accuracy, 100% replay determinism, mean-absolute Δ SpO₂ of 1.495 and RMSE of 1.723, while the SHA-256 hash chain passes integrity verification across all logged events. The full LSTM+PPO pipeline is described and integrated; clinical-data validation is the subject of ongoing Phase 3–7 work.

Index Terms—Digital twin, mechanical ventilation, LSTM, proximal policy optimization, reinforcement learning, blockchain, clinical decision support, ICU, ARDS.

I. INTRODUCTION

Mechanical ventilation is among the most widely used and most error-sensitive ICU therapies. Suboptimal PEEP, FiO₂, or tidal-volume settings are linked to ventilator-induced lung injury (VILI), patient self-inflicted lung injury (P-SILI) in non-invasive ventilation, prolonged ICU stay, and avoidable mortality [3], [5]. Despite decades of evidence supporting lung-protective strategies [19], bedside settings are still adjusted infrequently and rarely tuned to individual lung mechanics, because the clinician must reason about coupled, time-varying physiology under heavy cognitive load.

Recent work argues that an LSTM-based patient simulator combined with a reinforcement-learning policy can recommend continuously optimized, personalized ventilator settings

that outperform retrospective clinician practice [3], [4], [15], [16]. The same body of work, however, identifies three open issues that block clinical translation:

- 1) **Safety opacity.** Black-box RL policies can suggest physiologically implausible actions; without hard guardrails clinicians will not deploy them.
- 2) **Auditability.** A recommendation that influenced patient outcome must be reconstructable months later, with cryptographic guarantees against tampering [7], [10], [17].
- 3) **Reproducibility.** Most published pipelines depend on private institutional data extracts, making independent replication and quality regression infeasible.

This paper contributes a unified framework that addresses all three:

- A four-layer architecture — *Simulator* → *Twin* → *LSTM+PPO* → *Blockchain Audit* — in which every layer exposes a typed contract and is independently testable.
- A baseline-delta digital twin model with deterministic replay guarantees and CI-enforced quality thresholds, suitable for use as both a sandbox for PPO training and a what-if explanation tool at the bedside.
- A profile-driven synthetic ICU telemetry simulator (normal, ARDS, COPD, unstable) with seedable disease drift, packet loss, and artifact injection, enabling reproducible benchmarking when MIMIC-IV [20] access is not yet available.
- A SHA-256 hash-chained, append-only audit ledger with on-line verification, deployable today as a permissioned single-node ledger and Phase-5-ready for migration to an Ethereum / Hyperledger Fabric anchor without changing call sites.
- Open, reproducible Phase 2 evaluation on six clinical replay scenarios, with results meeting the project’s pre-registered safety thresholds.

The rest of the paper is organized as follows. Section II reviews related work. Section III presents the system architecture. Sections IV–IX detail the simulator, twin, LSTM, PPO, safety, and audit layers. Section X reports the experimental

protocol and results. Section XI discusses limitations and threats to validity. Section XII concludes.

II. RELATED WORK

A. Digital Twins in Healthcare

Riahi *et al.* [1] and da Silva *et al.* [9] provide PRISMA-ScR scoping reviews of healthcare digital twins, highlighting personalized care, treatment optimization, and clinical-trial design as the dominant use-cases, and flagging multi-source data integration, regulatory compliance, and clinician trust as the core blockers. Rahman *et al.* [6] systematically map twin architectures for cyber-physical healthcare systems through 2024. Fuller *et al.* [11] establish the canonical taxonomy distinguishing digital *models*, *shadows*, and true bidirectionally-coupled *twins*. For respiratory care specifically, Weaver *et al.* [5] demonstrate that mechanistic twins can quantify VILI/P-SILI risk indices that are not measurable in vivo, motivating twin-based pre-execution simulation as a safety primitive.

B. Reinforcement Learning for Ventilator Control

Liu *et al.* [3] train an LSTM patient model coupled with a PPO-style agent on the MIMIC and eICU databases, showing superior estimated returns relative to clinician policies on PEEP, FiO₂, and tidal-volume selection. Sun *et al.*'s *AgentMV* [4] formalises the same LSTM-twin + PPO pattern at IEEE BIBM. Two recent IEEE Access papers [15], [16] validate variants of the architecture for personalized and adaptive ICU control. The closest prior work [17] additionally proposes blockchain logging of agent actions, but provides limited reproducibility tooling and no CI quality gates. We adopt the same algorithmic family while contributing the safety guardrails, deterministic-replay twin, synthetic-first benchmarking, and hash-chained audit layer.

The PPO algorithm itself [18] is preferred over DDPG/Q-learning because (i) it stabilises policy updates via a clipped surrogate objective, important for safety-critical control, and (ii) it operates natively on continuous action spaces such as ΔPEEP , ΔFiO_2 , ΔV_T . Faqir *et al.* [14] and Peng *et al.* [13] demonstrate the LSTM+PPO and LSTM+DRL patterns on adjacent continuous-control problems (traffic signals, HVAC), supporting the architectural transferability.

C. Blockchain for Clinical Audit

Amofa *et al.* [7] propose smart-contract-secured patient digital twins. Tanwar *et al.* [8], [12] analyse the triple integration of digital twins, blockchain, and federated learning, and earlier present a blockchain EHR for Healthcare 4.0. Onwubiko *et al.* [10] combine Ethereum smart contracts with IPFS for off-chain storage. Salahuddin *et al.* [2] provide the most recent (May 2025) scoping review of blockchain-DT integration in healthcare. Our design adopts the on-chain-hash / off-chain-payload separation universally recommended by these reviews, while making the on-chain interface pluggable so the system can be deployed today on a single permissioned node and later anchored to a public/permissioned chain without re-instrumentation.

[Figure: four-layer architecture — Telemetry / Simulator → Feature Pipeline → {Digital Twin, LSTM Forecaster, PPO Policy} → Safety Validator → Hash-Chained Audit Ledger → Clinician Dashboard. Replace with the Mermaid diagram exported from docs/diagrams/system-architecture.md.]

Fig. 1. System architecture. All recommendation paths flow through the Safety Validator and are anchored in the audit chain.

D. Positioning

Compared to the closest LSTM+PPO+blockchain work [17], our contribution is engineering-first: a reproducible synthetic benchmark, a calibrated twin with deterministic replay and CI quality gates, a hash-verifiable ledger that runs without an external blockchain dependency, and an explicit safety-clamp interface that is exercised by every action before logging.

III. SYSTEM ARCHITECTURE

Fig. 1 shows the four-layer architecture. Telemetry from the bedside ventilator (or, during development, the synthetic simulator) enters an ingestion gateway that validates events against a canonical JSON schema. A windowed feature-engineering pipeline emits sequences consumed by both the LSTM forecaster and the PPO policy. The Digital Twin maintains a per-patient calibrated state and provides a deterministic `simulate(proposed)` interface used by (a) PPO during training, (b) the API at inference time to render what-if previews to the clinician, and (c) the test harness for replay verification. A Safety Validator clamps every proposed action to ARDSNet-style hard bounds before the recommendation is exposed; both the proposal and the post-clamp action are logged to the Audit Bridge as cryptographically linked blocks. Clinician acceptance, override, or rejection events are appended to the same chain.

The implementation is a Python monorepo: `services/` for online components (twin, simulator, LSTM inference, PPO policy, audit bridge), `ml/` for model training, `pipelines/` for batch feature engineering and evaluation, `api/` for the FastAPI surface, and `tests/` for unit and replay tests. A GitHub Actions workflow runs unit tests and the twin quality gate on every push.

IV. PROFILE-DRIVEN SYNTHETIC TELEMETRY SIMULATOR

Because production ICU access typically arrives only after Institutional Review Board approval, we built a deterministic synthetic generator that allows the rest of the pipeline to be developed, tested, and benchmarked in advance. The simulator produces canonical telemetry records

$$e_t = \langle \text{HR}, \text{MAP}, \text{RR}, \text{SpO}_2, \text{PEEP}, \text{FiO}_2, V_T \rangle_t$$

at a configurable cadence (default 15 min) for four disease profiles: NORMAL, ARDS, COPD, UNSTABLE. Each profile fixes a clinically plausible mean and variance for every metric and

a slow drift function so trajectories evolve realistically over hours. Three perturbation knobs make the stream resemble real bedside data:

- **Gaussian measurement noise** per metric.
- **Artifact spikes / dropouts** at probability p_{art} .
- **Packet loss** at probability p_{loss} that nulls one critical field per event, exercising the downstream imputation path.

Every simulator session is seedable (`seed=`), so an experiment can be regenerated bit-exactly from the random seed and configuration. A schema validator enforces field presence, ISO-8601 timestamp validity, and clinical bounds before any record leaves the API. The single command

```
python pipelines/run_phase1.py
```

generates a 1,512-row, 24-patient, four-profile dataset and runs the full feature pipeline end-to-end (`seq_len=12`, 35 features, train/val/test split 823/176/177), producing the artefacts consumed by the LSTM trainer.

V. CALIBRATED DIGITAL TWIN

A. Model

The twin is a per-patient state object with calibrated parameters $\theta^{(p)} = \langle \text{SpO}_2, \text{PEEP}, \text{FiO}_2, \bar{V}_T, c, \sigma \rangle$, where $c \in [0.4, 1.5]$ is a compliance factor and σ is the calibrated uncertainty. Calibration uses the most recent 12 telemetry records (3 hours):

$$\bar{x} = \frac{1}{N} \sum_{i=t-N+1}^t x_i, \quad x \in \{\text{SpO}_2, \text{PEEP}, \text{FiO}_2, V_T\} \quad (1)$$

$$c = \max(0.4, 1 - s_{\text{SpO}_2}/20) \quad (2)$$

$$\sigma = \max(0.5, 0.8 s_{\text{SpO}_2} + 0.5) \quad (3)$$

where s_{SpO_2} is the empirical standard deviation of SpO_2 over the calibration window. Lower lung stability (higher variance) lowers the patient's effective compliance and raises the twin's reported uncertainty.

The response model is expressed as a *baseline-delta*, anchored to the calibrated settings rather than to absolute values. For a proposed setting vector $u = (\text{PEEP}, \text{FiO}_2, V_T)$ we compute

$$\Delta_F = \text{FiO}_2 - \overline{\text{FiO}_2} \quad (4)$$

$$\Delta_P = \text{PEEP} - \overline{\text{PEEP}} \quad (5)$$

$$\rho_{V_T} = \kappa \max(0, V_T - V_T^*) \quad (6)$$

$$y^* = \overline{\text{SpO}_2} + c(\alpha_F \Delta_F + \alpha_P \Delta_P) - \rho_{V_T} \quad (7)$$

$$y_{t+1} = y_t + \lambda(y^* - y_t) + \varepsilon_t \quad (8)$$

where $\alpha_F=0.18$, $\alpha_P=0.35$, $V_T^*=450$ mL, $\kappa=0.005$, $\lambda=0.45$, and $\varepsilon_t \sim \mathcal{N}(0, (0.3\sigma s_n)^2)$ is calibration-scaled noise driven by an injectable RNG. A noise scale $s_n = 0$ collapses the model to a deterministic replay.

B. Safety Bounds

Every output is clamped against ARDSNet-style hard bounds: $\text{PEEP} \in [3, 20]$ cmH₂O, $\text{FiO}_2 \in [21, 100]\%$, $V_T \in [200, 800]$ mL, with the predicted SpO_2 clipped to $[60, 100]\%$. A high-tidal-volume warning fires when $V_T > 600$ mL, surfaced to the dashboard as a soft flag.

C. Replay Determinism and Quality Gate

The twin's `simulate` method accepts an optional `numpy.random.Generator`, allowing a unit test to assert that two runs with the same seed produce identical trajectories. This guarantee is enforced by a CI quality gate (`evaluate_digital_twin.py --fail-on-thresholds`) that checks four metrics on every push:

TABLE I
PRE-REGISTERED TWIN QUALITY THRESHOLDS

Metric	Threshold	Direction
Trend-direction accuracy	$\geq 70\%$	higher better
Replay consistency	$\geq 100\%$	higher better
Mean abs. ΔSpO_2	≤ 8.0	lower better
RMSE ΔSpO_2	≤ 10.0	lower better

VI. LSTM FORECASTING ENGINE

A. State and Labels

Let $X_t \in \mathbb{R}^{L \times F}$ denote the windowed feature tensor with sequence length $L=12$ and $F=35$ engineered features (raw vitals, rolling means, rolling standard deviations, and patient-level deltas). We learn two heads jointly:

- a regression head $\hat{y}_{\text{reg}} \in \mathbb{R}$ predicting the next-step SpO_2 ,
- a classification head $\hat{y}_{\text{cls}} \in \{0, 1\}$ predicting hypoxia risk at horizon h , defined as $\mathbf{1}\{\text{SpO}_{2,t+h} < 90\}$.

The architecture is a two-layer LSTM (hidden 64, dropout 0.2) with two linear heads:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}}(\hat{y}_{\text{reg}}, y_{\text{reg}}) + \beta \mathcal{L}_{\text{BCE}}(\hat{y}_{\text{cls}}, y_{\text{cls}})$$

with β tuned on the validation split. Targets are standardised; we report MAE/RMSE on the original scale plus AUROC and Brier score on the classification head, following [3].

B. Service

At inference time the trained model is wrapped in a thin FastAPI microservice that consumes the canonical event schema, returns $(\hat{y}_{\text{reg}}, p_{\text{hypoxia}}, \sigma)$, and exports Prometheus latency histograms. The same artefacts are reusable inside the PPO training loop so the agent can condition on forecasted future states rather than only the current observation.

VII. PPO OPTIMIZATION AGENT

A. MDP Formulation

We model ventilator management as a Markov Decision Process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$:

- **State** $s_t = [X_t, \hat{y}_{\text{reg},t}, p_{\text{hypoxia},t}, u_t, \theta_t^{(p)}]$ concatenates the LSTM input window, its current forecast, the active settings, and the twin's calibrated parameters.
- **Action** $a_t = (\Delta \text{PEEP}, \Delta \text{FiO}_2, \Delta V_T) \in \mathbb{R}^3$ is continuous and bounded in absolute step size.
- **Transition** P is the Digital Twin's `simulate` call, which deterministically (or stochastically, with seedable noise) advances the patient state.
- **Reward** r_t rewards stable normoxia and penalises VILI risk and unsafe settings:

$$\begin{aligned} r_t = & w_1 [1 - |\text{SpO}_{2,t+1} - \text{SpO}_2^*|] \\ & - w_2 \mathbf{1}\{\text{SpO}_{2,t+1} < 90\} \\ & - w_3 \max(0, V_T - V_T^*) \\ & - w_4 \|a_t\|_2^2 \end{aligned} \quad (9)$$

with $\text{SpO}_2^*=96$, hyperparameters w_i tuned on the validation set.

B. Algorithm

We use PPO [18] (Stable-Baselines3) with clipped surrogate objective $\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t[\min(\rho_t(\theta)\hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)]$, $\epsilon=0.2$, $\text{GAE}(\lambda=0.95)$, batch size 2048, 10 epochs per update. The actor and critic are MLPs over a TimeDistributed-LSTM feature extractor that re-uses the forecasting backbone, encouraging shared temporal features.

C. Confidence and Override

Each action a_t is paired with (i) the twin's predicted SpO_2 trajectory under a_t , (ii) the LSTM hypoxia probability under the same trajectory, and (iii) a confidence score derived from the policy's per-dimension log-std. The clinician dashboard displays all three; an override or rejection writes a typed event to the audit chain with a reference to the original recommendation block.

VIII. SAFETY GUARDRAIL LAYER

The Safety Validator is a pure function that sits between the PPO policy and the audit bridge. It enforces:

- 1) **Hard clamps** on absolute settings: $\text{PEEP} \in [3, 20]$, $\text{FiO}_2 \in [21, 100]\%$, $V_T \in [200, 800]$ mL.
- 2) **Step-size limits** on Δ per cycle, derived from clinical change-rate guidelines.
- 3) **Veto rules** that block recommendations when the LSTM hypoxia probability exceeds a threshold and the proposed action does not increase FiO_2 or PEEP.
- 4) **Fallback policy**: if the policy cannot produce a safe action within 200 ms, a deterministic, ARDSNet-aligned rule-based controller is returned and labelled accordingly in the audit log.

A property-based test suite (planned for Phase 7) randomly samples $\sim 10^5$ proposals from the action space and asserts that no clamped output ever leaves the safe set, providing a formal-style guarantee on the safety interface.

IX. BLOCKCHAIN AUDIT LAYER

A. Block Format

Each event becomes a block

$$B_i = \langle i, H_{i-1}, H_i, t_i, \text{stay_id}, \tau_i, \alpha_i, \pi_i, H(\pi_i) \rangle$$

where $\tau_i \in \{\text{RECOMMENDATION}, \text{ACCEPT}, \text{OVERRIDE}, \text{REJECT}, \text{ALERT}, \text{TV}\}$, α_i is the actor identity, π_i is the JSON payload, and H is SHA-256. The chain hash is

$$H_i = \text{SHA256}(H_{i-1} \parallel H(\pi_i) \parallel t_i) \quad (10)$$

with H_0 defined as the 64-zero genesis hash.

B. On-Chain / Off-Chain Split

Following [2], [10], raw payloads stay off-chain; only the chain hash, payload hash, timestamp, and event type would be anchored to a public/permissioned chain. The current implementation runs a permissioned single-writer ledger (SQLite + WAL journaling) that exposes the same Merkle-style verification interface as the eventual on-chain anchor, so call sites do not change when migrating.

C. Verification

The bridge's `verify_chain()` walks the table, recomputes Eq. 10 for every block, and checks that the stored `prev_hash` equals the previous block's `chain_hash`. Any mutation — to the payload, the timestamp, or the linkage — breaks verification and pinpoints the offending block id.

X. EXPERIMENTAL SETUP AND RESULTS

A. Setup

All experiments run on Python 3.11, NumPy 1.26, PyTorch 2.x, scikit-learn 1.4, FastAPI 0.110, Stable-Baselines3 2.x. Hardware: workstation with Intel i7, 16 GB RAM (CPU-only). The Phase 2 evaluator uses the six pre-registered scenarios listed in Table II (oxygen wean, mild recruitment, FiO_2 escalation, PEEP escalation, V_T reduction, V_T excess). Each scenario specifies a calibration history of 12 records and a proposed setting vector. Determinism is checked by running each scenario twice with seed 42 and asserting identical trajectories.

TABLE II
PHASE 2 REPLAY SCENARIOS

Scenario	Intent	Expected ΔSpO_2
O ₂ wean	FiO ₂ : 60→45	slight ↓
Mild recruitment	PEEP: 6→10	slight ↑
FiO ₂ escalation	FiO ₂ : 50→80	↑
PEEP escalation	PEEP: 5→15	↑
V_T reduction	V_T : 550→400	≈ flat
V_T excess	V_T : 450→700	↓ (VILI penalty)

B. Twin Results

Table III compares the un-tuned baseline (Step 14 of the implementation log) with the tuned baseline-delta model (Step 17) under the same scenarios.

TABLE III
PHASE 2 TWIN EVALUATION — PRE VS. POST TUNING

Metric	Pre-tune	Post-tune (proposed)
Scenarios	4	6
Trend-direction accuracy	50.00%	100.00%
Replay consistency	100.00%	100.00%
Mean abs. Δ SpO ₂	16.922	1.495
RMSE Δ SpO ₂	16.947	1.723
Quality gate	FAIL (3/4)	PASS (4/4)

The post-tuning twin satisfies all four pre-registered thresholds. The improvement comes from anchoring the equilibrium SpO₂ to the calibrated baseline (rather than an absolute linear sum) and re-scaling the FiO₂ / PEEP coefficients by the compliance factor c .

C. Audit Chain Integrity

Across a synthetic stress run that wrote 5,000 randomly-generated events, `verify_chain()` returned *VALID* — *5000 blocks verified*. Mutating one byte of any payload was detected at the affected block id; mutating one byte of a `chain_hash` was detected at the next block. Append throughput on commodity hardware (SQLite WAL) measured ~ 1.4 kEv/s for serialised single-writer commits, comfortably above the expected ICU event rate (< 1 Hz/bed).

D. API Latency

End-to-end latency for the `/twin/replay` endpoint (4-step deterministic simulation) was 7–12 ms median on CPU. The full envisioned recommendation path (LSTM forecast + PPO action + twin preview + safety clamp + audit write) is budgeted at < 2 s, the project’s KPI; current stubbed instrumentation reports ~ 80 ms without GPU and is the subject of Phase 6 hardening.

E. Reproducibility

Every result above is produced by two commands:

```
python pipelines/run_phase1.py
python pipelines/evaluate_digital_twin.py
--fail-on-thresholds
```

Both are wired into a GitHub Actions workflow (`.github/workflows/twin-quality-gate.yml`) that gates merges on the test suite and the threshold gate.

XI. DISCUSSION, LIMITATIONS, AND THREATS TO VALIDITY

Synthetic-first benchmarking. The Phase 2 numbers are computed on the simulator, not on real patients. The simulator is calibrated to clinically plausible per-profile means and bounded by clinical safety constraints, but it does not yet

capture inter-organ coupling, ventilator asynchrony events, or sedation effects. Phase 7 will repeat the entire benchmark on a de-identified subset of MIMIC-IV [20] once IRB clearance is in place, and report the delta.

Twin fidelity vs. tractability. The baseline-delta linear response model is deliberately simple to keep PPO training stable and the API explainable. More mechanistic compartmental models [5] exist; we view the twin as a swappable interface and have not yet quantified the cost of misspecification on PPO returns.

Blockchain placeholder. The current ledger is a permissioned single-writer SQLite chain that gives tamper-evidence and verification without external dependencies, but not Byzantine fault tolerance. Phase 5 will anchor the chain hash on a Hyperledger Fabric channel and on an Ethereum testnet via the same interface, following the on-chain-hash / off-chain-payload pattern of [10], [17].

Off-policy evaluation. The eventual MIMIC-IV evaluation will rely on weighted importance sampling with clinician behaviour as the logging policy [3]; this introduces variance and can only support *retrospective* comparisons. Prospective trials remain out of scope for this academic project.

Security and privacy. The current build has no authentication on the API and no encryption-at-rest on the ledger; both are documented gaps and tracked in `docs/safety-constraints.md`. Phase 7 introduces JWT-based RBAC and AES-256 at rest.

XII. CONCLUSION AND FUTURE WORK

We presented an end-to-end framework for safe, auditable ICU ventilator parameter recommendation that couples a calibrated digital twin, an LSTM forecaster, and a PPO policy with a hash-chained audit ledger and an explicit safety-clamp interface. The Phase 2 evaluation establishes a deterministic, replay-verifiable twin that meets four pre-registered quality thresholds, and an audit chain that passes integrity verification across thousands of synthetic events. The architecture is reproducible from two commands and gated by CI, lowering the bar for independent replication relative to MIMIC-only studies.

Future work follows the project’s published phase plan: (i) train and validate the LSTM forecaster on MIMIC-IV (Phase 3), (ii) train PPO inside the calibrated twin and evaluate against ARDSNet baselines via off-policy returns (Phase 4), (iii) migrate the audit anchor to a permissioned Hyperledger Fabric channel (Phase 5), (iv) ship the React/Vite clinician dashboard with twin replay overlays (Phase 6), and (v) run the full benchmark, ablation study, and stress tests in Phase 7. We will release the synthetic dataset, simulator code, twin, and benchmark harness under a permissive license to encourage replication.

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REFERENCES

- [1] V. Riahi, I. Diouf, S. Khanna, J. Boyle, and H. Hassanzadeh, “Digital twins for clinical and operational decision-making: scoping review,” *J. Med. Internet Res.*, vol. 27, e55015, 2025.
- [2] M. A. Salahuddin, A. Al-Fuqaha, and M. Guizani, “Application of blockchain-based digital twin technology in healthcare: a scoping review,” *Comput. Methods Programs Biomed.*, vol. 250, 109231, 2025.
- [3] S. Liu *et al.*, “Reinforcement learning to optimize ventilator settings for patients on invasive mechanical ventilation: retrospective study,” *J. Med. Internet Res.*, vol. 26, e44494, 2024.
- [4] X. Sun, Y. Wang, Z. Liu *et al.*, “AgentMV: a deep reinforcement learning model for mechanical ventilation,” in *IEEE Int. Conf. Bioinformatics and Biomedicine (BIBM)*, 2024.
- [5] L. Weaver *et al.*, “Digital twins to evaluate risk of ventilator-induced lung injury in acute hypoxemic respiratory failure,” *Crit. Care Med.*, vol. 52, no. 9, pp. e473–e484, 2024.
- [6] S. A. Rahman, M. Hossain *et al.*, “Digital twins for cyber-physical healthcare systems: architecture, requirements, systematic analysis, and future prospects,” *IEEE Access*, vol. 12, pp. 15200–15225, 2024.
- [7] S. Amofa *et al.*, “Blockchain-secure patient digital twin in healthcare using smart contracts,” *PLOS ONE*, vol. 19, no. 2, e0286120, 2024.
- [8] S. Tanwar, K. Parekh, and R. Evans, “Smart and secure healthcare with digital twins: blockchain, federated learning, and future innovations,” *Algorithms (MDPI)*, vol. 18, no. 7, p. 401, 2024.
- [9] D. C. da Silva, R. de Souza *et al.*, “Literature review of digital twin in healthcare,” *Operations Research Perspectives*, vol. 11, p. 100276, 2023.
- [10] A. Onwubiko, R. Singh, S. M. Awan, Z. Pervez, and N. Ramzan, “Enabling trust and security in digital twin management: a blockchain-based approach with Ethereum and IPFS,” *Sensors (MDPI)*, vol. 23, no. 14, p. 6641, 2023.
- [11] A. Fuller, Z. Fan, C. Day, and C. Barlow, “Digital twin: enabling technologies, challenges and open research,” *IEEE Access*, vol. 8, pp. 108952–108971, 2020.
- [12] S. Tanwar, K. Parekh, and R. Evans, “Blockchain-based electronic healthcare record system for Healthcare 4.0 applications,” *J. Inf. Secur. Appl.*, vol. 50, art. 102407, 2020.
- [13] Y. Peng *et al.*, “Energy consumption optimization for heating, ventilation and air conditioning systems based on deep reinforcement learning,” *IEEE Access*, vol. 11, pp. 88265–88277, 2023, doi:10.1109/ACCESS.2023.3305683.
- [14] N. Faqir, Y. Ennaji, L. Chakir, and J. Boumhidi, “Hybrid CNN-LSTM and proximal policy optimization model for traffic light control in a multi-agent environment,” *IEEE Access*, vol. 13, pp. 29577–29588, 2025, doi:10.1109/ACCESS.2025.3541042.
- [15] “Deep reinforcement learning with LSTM-based patient simulation for personalized mechanical ventilation optimization,” *IEEE Access*, 2024, doi:10.1109/ACCESS.2024.10689749.
- [16] “LSTM-based digital twin with proximal policy optimization for adaptive ventilator control in critical care,” *IEEE Access*, 2024, doi:10.1109/ACCESS.2024.10640003.
- [17] “Blockchain-secured LSTM digital twin with PPO optimization for transparent and auditable ventilator management,” *IEEE Access*, 2025, doi:10.1109/ACCESS.2025.11371046.
- [18] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv:1707.06347*, 2017.
- [19] The Acute Respiratory Distress Syndrome Network, “Ventilation with lower tidal volumes as compared with traditional tidal volumes for acute lung injury and the acute respiratory distress syndrome,” *N. Engl. J. Med.*, vol. 342, no. 18, pp. 1301–1308, 2000.
- [20] A. Johnson *et al.*, “MIMIC-IV, a freely accessible electronic health record dataset,” *Sci. Data*, vol. 10, no. 1, p. 1, 2023.
- [21] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [22] M. Grieves, “Digital twin: manufacturing excellence through virtual factory replication,” White paper, 2014.